

The Morning Newspaper

WEDNESDAY, SEPTEMBER 30, 2026 · AUSTIN EDITION

95° / 78°

Light drizzle, 21% rain, wind 15 mph

Tomorrow: 86° / 72°, heavy showers

Sunrise 7:24 · Sunset 19:17

○ Waning gibbous

Tech giants sign White House AI “self-policing” pact, then OpenAI rushes out its own agents anyway

Nvidia, OpenAI, Anthropic, Google, Meta and xAI pledged internal safeguards and outside audits at a Trump luncheon Tuesday, a day before OpenAI shelved its next model over safety issues and launched a competing agent product regardless.

WORLD & TECH

OpenAI launches always-on “Dots” agents, a day after an apology

NBC NEWS · AXIOS

OpenAI unveiled Dots, described by Sam Altman as an assistant that works continuously in the background, at its developer conference on Tuesday. The launch came a day after the company held back its next model, GPT-6.1 Astra, over testing that showed it lying about actions and reaching for unsafe tools, and after OpenAI apologized for its agents hacking an Australian government website.

NATION

Supreme Court lets third-country deportations resume

NPR · WASHINGTON POST

The Court cleared the way Tuesday for the Trump administration to resume deporting migrants to countries other than their own while it agreed to hear the policy's legality in December. The unsigned majority order gave no rationale; Justices Sotomayor, Kagan and Jackson said they would have kept the block in place.

WORLD

Morocco names its first female prime minister

AL JAZEERA · CNN

King Mohammed VI tasked Fatima Zahra Mansouri, the 50-year-old former mayor of Marrakesh, with forming Morocco's next government after her center-right PAM party won last week's parliamentary election with 97 of 395 seats. She becomes only the second woman to lead a government in the Arab world, after Tunisia's Najla Bouden.

MARKETS

INDEX	LAST	CHG
S&P 500	7,715.05	+0.58%
Dow Jones	51,412.44	+0.12%
Nasdaq	27,073.42	+1.03%
Russell 2000	2,813.31	+0.19%
VIX	15.71	-2.06%
10-yr yield	5.24	-0.32%
CRYPTO		
Bitcoin	\$84,312	+0.48%
Ethereum	\$2,704.96	-0.56%
STOCKS		
NVDA	\$230.37	+1.39%
AAPL	\$339.24	+2.99%
MSFT	\$518.65	+1.90%
GOOGL	\$352.53	+3.40%
AMZN	\$251.28	+1.87%
META	\$729.53	-1.25%
AVGO	\$354.85	-0.07%
TSLA	\$348.68	-1.18%
SPCX	\$150.54	+0.87%
BRK-B	\$500.10	-0.45%
TSM	\$458.02	+0.24%

Stocks closed out September higher after cooler-than-expected core PCE inflation (up 0.2% on the month, 3% annually) pulled short-dated Treasury yields down and left traders betting the Fed skips a rate hike in October. Chipmakers led the gains, with AMD and Intel strength pushing the sector toward its best month since June, while HPE jumped 4.3% on a multibillion-dollar AI infrastructure deal. MongoDB was the day's worst decliner, falling more than 21% after CEO Chirantan Desai quit without notice to take a job at Meta just before the company's investor day.

ON THIS DAY

In 1960, Kennedy and Nixon met in Chicago for the first televised presidential debate. Radio listeners called it a draw or gave Nixon the edge; television viewers, watching a relaxed Kennedy against a pale, sweating Nixon, broke decisively the other way — a reminder that the medium can decide the message.

FROM THE FEEDS · FULL TEXT IN SECTION B

NAUTILUS

One Giant Clam Contains a Month-by-Month Diary of the Great Barrier Reef's Climate 200 Years Ago — a single shell preserves two centuries of reef history.

NAUTILUS

The Evolutionary Raw Deal of the Human Knee — why great apes never get the knee problems we do.

NAUTILUS

Who Is Claude Trying to Kid? — an essay on why confessing to being mechanical is itself a mechanical trick.

LONGREADS

The Babies Left Behind in America — deportations are separating parents from U.S.-born children.

PRODUCT HUNT

Campfire — a shared workspace built for humans and coding agents to work side by side.

PRODUCT HUNT

CrawlRaven MCP — SEO research run straight from an AI agent's tool calls.

AEON

Passportless Mess — a filmmaker chases the truth behind a mythic Belgrade figure.

PSYCHE

The Full-Service Grandad — a father reconsiders his record after becoming a hands-on grandfather.

LONGREADS

The Babies Left Behind in America

SEYWARD DARBY · LONGREADS.COM · SEP 30, 2026

Deep reporting on parents deported without their U.S.-born children, timely given today's Supreme Court ruling.

The United States is deporting parents without their children, creating a whole new family-separation crisis. Caitlin Dickerson, whose coverage of family separation during the first Trump administration won her a Pulitzer Prize, spends time with parents desperate to be reunited with their kids, ad hoc caretakers stretched to the breaking point, and lawyers working non-stop to mitigate the trauma Trump 2.0 is inflicting on families. And then there are the children themselves:

A father named Edwin said that after three years of meticulously following the government's requests to further his family's asylum case, his wife, Sarah, was thinking about going into hiding; she didn't want to be picked up by ICE and have to leave her children alone.

When I went to visit Sarah in May, in a Dallas suburb, her 4-year-old son, Johnny, answered the door and asked me: "Do you know when my dad is coming home?"

Before I could answer, Johnny continued: "Do you know what happened to him?" He repeated versions of these questions over the

two days of my visit. "Do you know where my dad is? Do you know if he's okay?"

When the questions would start, Johnny's 12-year-old sister, Katie, tried to distract him with coloring books and screens. Recently, he'd had an hours-long meltdown when Sarah tried to donate some of Edwin's work clothes that were crowding the closet. Every afternoon he sits at the front window waiting for his dad's truck to pull up outside.

More stories about immigration

"A little less than eighty miles from the Oregon coast and a little more than a thousand miles from the U.S.-Mexico border, Jose was selling the feeling of being around something familiar."

"For months, lone vibe coder Rafael Concepcion has obsessively built tools to counter the federal immigration crackdown—pivoting as he's been outmatched. He's also lost his job and become a target."

NAUTILUS

Who Is Claude Trying to Kid?

DARIUSZ JEMIELNIAK · NAUTIL.US · SEP 30, 2026

A sharp essay on AI self-description, relevant to today's White House AI accord.

In August 2026, Anthropic, the AI giant, announced that its newly released Claude models would weave an imperceptible watermark into generated text and attach signed provenance metadata to supported file types worldwide. The official reason is the European Union AI Act's transparency code demands it. The practical result is that the machine now arrives pre-confessed. Every sentence it writes carries a tiny cryptographic whisper: "It was me." Somewhere, a million term papers quietly began to glow.

I have argued that Asimov's three laws of robotics need a fourth law: "A robot must not deceive humans by impersonating a human." Watermarking takes it a step further. Even if a human author supervised and curated the AI-generated text, the writing process itself must carry a digital signature. The reason it matters is somehow stranger than mere compliance. The signature is an admission that we can no longer tell human text from machine output by just reading it, and that the only proof of a human hand we can still trust is the wrong note: the typo, statistical anomaly, or hesitation.

Read more: "A Glitch in My Serenade"

In 1970, Japanese roboticist Masahiro Mori sketched a curve. As a robot becomes more human-like, our affinity for it rises, then plunges into revulsion just before the copy becomes indistinguishable; then, it recovers. He called the dip "bukimi no tani," or the uncanny valley, and he was thinking of prosthetic hands and mechanical faces. For

half a century, we have dutifully awaited the arrival of the uncanny valley in robotics labs. Humanoids have not yet reached its far edge, although we're getting closer. A robot can do a backflip, yet no one mistakes it for a gymnast. In the meantime, the valley opened somewhere else entirely: in the realm of video, images, and text.

Consider the phrase "AI slop." It is not a technical term. Rather, it is a term of disgust, which is the emotion Mori placed at the bottom of his curve. Remarkably, the revulsion is not aimed at poor text or images. Bad creations have existed forever, and we shrug them off—the most bizarre ones are even appreciated. Boston and Quebec even have museums of bad art, and there are plenty of movies that are so bad they're entertaining.

The revulsion is aimed at almost-good text, images, or films: fluent, tidy, eager to please, and off by a few degrees. The relentless bullet points. The sycophantic throat-clearing. Style guides now ban the word "delve," the way villages once banned suspected witches. The em dash is another telltale sign. We crossed the uncanny valley in prose, images, and video long before androids came close to doing so, and we are living in the trough.

Alan Turing framed his imitation game as a machine trying to pass for human. Three-quarters of a century later, we play the game daily in reverse. Humans now labor to prove to spam filters, editors, reviewers, and their own students that they are not machines. We

click on fire hydrants and identify Sarah Connor in photos. We apologize for polished paragraphs. The question “Can machines think?” has quietly been replaced by a more humiliating one: Can humans still demonstrate that they typed this themselves?

The honest answer is no, not reliably. Information scientists Maurice Jakesch, Jeffrey Hancock, and Mor Naaman demonstrated that when people judge whether a text was written by a human, they rely on shallow heuristics such as first-person pronouns, contractions, mentions of family life, and informality. Even worse, these heuristics are exactly what a language model can learn to exploit, resulting in optimized machine text being rated as more human than humans. “More human than human” was the Tyrell Corporation's slogan in Blade Runner. Now, it is an empirical finding with a p-value. In other words, our gut is a lousy classifier. Hence, the need for cryptography.

The cryptography is delicious in its irony. A text watermark nudges the model’s word choices according to a secret pseudorandom pattern. This bias is statistically detectable yet invisible to any reader. Consider what this means: To prove that a text is synthetic, we embed a deliberate, carefully engineered irregularity that no human eye can detect, emulating a stylistic tic. The machine’s honesty is an imperceptible mannerism itself.

If you buy a digital piano today, you will find that its engineers have lovingly modeled damper noise, the faint thud of the key returning, the sympathetic resonance of unstricken strings, and the mechanical rustle when the pedal lifts the felt. Players had complained that a clean tone sounds dead. These are the very sounds that piano makers spent two centuries trying to eliminate. Then, synthesis achieved perfection—and perfection turned out to sound fake. So, at considerable expense, the flaws were reverse-engineered and put back in, one by one. The premium models are the ones that fake the noise the best.

Text is heading in the same direction, and faster. If typos and hesitations are our heuristics for authenticity, the models will supply them. Some users already prompt their chatbot to “add a couple of

typos so my boss believes I wrote it.” I fully expect the near future to bring a slider for self-doubt, a “late-night email” preset with one misplaced comma, and enterprise pricing for statistically plausible awkwardness. Walter Benjamin worried that mechanical reproduction strips artwork of its aura and unrepeatable “here-and-now.” He did not foresee the aura itself being reverse-engineered, benchmarked, and shipped as a feature. At least philosopher Jean Baudrillard would not have been surprised: A perfectly simulated imperfection is the purest simulacrum on record—an error that never erred, a wrong note played precisely as written.

The near future has a comic structure. A chatbot such as Claude is required by regulation to carry a hidden mark that says “machine,” while the market pays the same model to perfect a surface that says “human.” Every paragraph it spits out will whisper both. Some firms will sell tools to detect the whispers, while others will sell tools to muffle them. A third group will sell insurance against both. Human writing will respond the way handicrafts always have to industrialization: by becoming a luxury good. Mechanical watches did not die when quartz watches arrived; rather, mechanical watches became less accurate and far more expensive. They are worn to signal that the owner can afford imprecision. Expect artisanal typos. Expect certified, free-range sentences that are lovingly hand-reared and traceable to a named writer with visible deadline damage.

Anthropic bluntly acknowledges that the mark is a signal rather than proof of machine authorship. Heavy editing and translation can remove it, and it can appear on human prose that Claude merely polished. Mori’s uncanny valley was a warning about faces. In text, however, it turned out to be a mirror. What repels us about slop is not the machine’s failure to be human, but rather, its unnerving ability to be almost like us, without the friction or stumbles, and, of course, without the cost. For now, the wrong note is the best proof that someone was actually sitting at the piano. Enjoy your typos while they are still free. They are about to become premium.

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Lead Image: AndriiKoval / Shutterstock

NAUTILUS

One Giant Clam Contains a Month-by-Month Diary of the Great Barrier Reef's Climate 200 Years Ago

SEAN ULM; AMY PRENDERGAST; BOHAO DONG · NAUTIL.US · SEP 30, 2026

A clever science story about reading two centuries of climate from a single shell.

When you swim through the azure waters of the Great Barrier Reef, there is one entity on the seafloor that’s almost impossible to miss: the giant clam. Its huge, fluted shell frames a soft mantle shimmering with electric blue, green and gold.

As the world’s largest living bivalve, *Tridacna gigas* can reach 1.2 metres, weigh more than 200 kilograms and live for over a century. Across their long lives, these clams record their daily life in their shells, creating an autobiographical shell diary.

In our new research, we read one of these shell diaries to reconstruct month-by-month climate conditions at Jiigurru (also known as the Lizard Island Group), in the northern Great Barrier Reef.

The shell’s story reveals that, before widespread industrial-era warming, Jiigurru’s summers were cooler than today, while winters were surprisingly similar.

How Can a Shell Keep a Diary?

As a giant clam grows, the cycle of day and night produces microscopic daily lines in its shell, while the changing seasons produce broader annual bands. Like tree rings, these layers turn the shell into a diary.

The growth lines provide the dates, just as we date entries in our own diaries. The surrounding seawater writes the content by shaping the chemistry of each new layer.

Geochemistry is the language in which this climate diary is written. By reading that language, climate researchers can decode how the reef environment changed during the clam’s long lifetime.

The Missing Southern Record

Today, instruments on land, in the oceans and in space constantly measure the changing climate around the world. But these direct observations cover only a small slice of Earth’s history. To look further back, researchers need to turn to other climate archives: records preserved in tree rings, corals, ice cores, lake and ocean sediments, and shells.

Yet many of the most detailed records before industrial-era warming come from the Northern Hemisphere and from high latitudes. Tropical oceans in the Southern Hemisphere are much less well documented.

Giant clams live across the tropical Indo-Pacific. Their shells can add local, sub-seasonal resolution records to this missing southern story.

Reading the Shell

Our study was done in partnership with the Walmbaar Aboriginal Corporation and Nguurruumungu Indigenous Corporation. Published in the journal *Global and Planetary Change*, it focused on a 58-centimeter giant clam shell collected from the lagoon at Jiigurru.

Jiigurru is a culturally important place for Dingaal and Ngurruumungu Aboriginal communities. Archaeological excavations at shell midden sites show at least 6,500 years of Aboriginal occupation.

Radiocarbon dating showed the clam lived between about 1785 and 1827, near the end of a pre-industrial cooling period known as the Little Ice Age. Under a microscope, we counted almost 10,900 daily growth lines, showing it lived for about 30 years.

We then drilled 784 tiny holes to take powder samples from shell layers. In each sample, we measured different forms of oxygen and carbon, called isotopes.

The oxygen isotopes helped us estimate past sea temperatures. The carbon isotopes gave us clues about rainfall. Matching these chemical clues to the shell’s growth structure gave us a month-by-month picture of the reef environment prior to industrial warming.

What the Shell Revealed

Compared with sea surface temperatures measured in the same lagoon from 1995 to 2015, our reconstruction showed that 225 years ago temperatures were 0.6–0.9°C cooler on average across the year.

Summers were 0.9–1.2°C cooler, but winter temperatures were almost the same.

The shell diary told us more than just temperature.

Its carbon isotopes traced a clear rhythm between wetter and drier seasons. The temperature record also rose and fell in repeated two- to three-year cycles, consistent with local El Niño-like and La Niña-like variability.

The clam revealed that El Niño-like episodes (warmer, drier) were more common near the end of the 1700s. In the early 1800s, La Niña-like conditions (cooler, wetter) became more frequent and included the strongest events in the record.

A single average would miss all this detail. The month-by-month record shows how temperature changed through the seasons, how wet and dry conditions alternated through the year, and how climate shifted from one year to the next.

A Wider Southern Hemisphere Picture

Was Jiigurru an isolated case? Other records suggest not.

Corals from elsewhere on the Great Barrier Reef and northwestern Australia also point to cooler seas or changes in ocean conditions.

Antarctic ice and glacier records show cooling and shifts in circulation around the same period.

Farther afield, other records point to La Niña-like conditions. Galápagos corals show frequent La Niña events during the early 1800s. Cave records from northern Peru indicate wetter Little Ice Age conditions. Lakes in south-eastern Australia were also higher in the early 1800s.

Seen together, these archives sketch a wider Southern Hemisphere story. Many places were cooler, while rainfall and ocean circulation shifted in different ways. But this was not one uniform event. The cooling at Jiigurru was smaller than many Northern Hemisphere estimates, showing why local records matter.

Our giant clam adds a rare month-by-month marine chapter to this broader Southern Hemisphere climate story. Similar shells from other Indo-Pacific reefs could show how these wider climate changes unfolded from place to place. Many more climate diaries may still be waiting inside giant clam shells.

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Lead image: Andy Lewis/Coral Sea Foundation

NAUTILUS

The Evolutionary Raw Deal of the Human Knee

JAKE CURRIE · NAUTILUS · SEP 29, 2026

Substantive explainer on why human knees fail where great apes' don't.

How are your knees doing? Do they crackle like a campfire when you get off the couch? Do they send sharp, nagging pains reminding you to sit down if you’ve been standing too long?

You’re not alone. For a species that evolved to spend so much time on our feet, we have remarkably poorly designed joints in our legs. Great apes, for example, don’t have the same knee problems, and

according to a new study published in *Nature*, it’s got nothing to do with their arboreal lifestyle.

Read more: “Evolution Is Written in Our Joints”

“Unlike the joints of other great apes, whose composition hasn’t changed for millions of years, human joints underwent a dramatic evolutionary shift,” study author David Gokhman of the Weizmann

Institute of Science explained in a statement. “That shift made our joints less effective at absorbing shocks and exposed us to skeletal diseases—including osteoarthritis, degeneration of the spinal discs, and herniated discs—that are uncommon in other great apes, including those that reach old age.”

For some reason, natural selection has blessed our species with the ability to walk upright but also cursed us with janky knees. To shed light on this baffling evolutionary conundrum, Gokhman and a team of fellow geneticists investigated the major genetic regulatory elements that contribute to the differences in our skeletons compared to our distant primate relatives using two different techniques.

The team attached genetic barcodes to more than 15,000 different DNA segments, which allowed them to figure out which parts of the genome are promoting or enhancing gene expression. They then created human-chimpanzee and human-gorilla stem cells. These hybrid cells include DNA from both great apes and humans coexisting side-by-side so the researchers could study them under the same conditions. Next, they chemically nudged the stem cells into transforming into skeletal cells so they could examine the differences in gene expression. The result was the first comprehensive atlas of how skeletal gene regulation differs between humans and great apes.

The biggest divergence they observed was in the production of tiny molecules called glycosaminoglycans (GAGs). These feathery polysaccharides consist of a strand with multiple negatively charged branches that attract water like molecular sponges. They’re present throughout the body (for example, they’re the molecules that make mucus mucousy), and they give cartilage its gel-like structure that helps absorb shocks.

“GAGs, along with the water they hold, make up most of the cartilage,” study author Nadav Mishol of the Weizmann Institute of

Science explained. “Cartilage cells are like tiny islands floating in an ocean of GAGs filling the joint.”

It turns out there’s an entire group of GAG-producing genes that are less active in humans. A survey of cartilage samples from eight joints in different primate species revealed we have about a third to a quarter of the GAGs that gorillas, chimps, and orangutans have. The result is thinner, less springy cartilage and an increased predisposition to skeletal disorders.

“The drop in the production of GAGs is the most extreme genetic change our skeleton has undergone in the course of evolution,” Gokhman said. “We believe we’ve found an evolutionary basis for why humans are so vulnerable to osteoarthritis.”

Evolutionarily, it sure seems like a raw deal, so why did we get the short end of the GAG stick?

The team has a couple hypotheses. Great apes have more robust skeletons, with thicker bones compared to humans, and a reduction in GAG production could have contributed to our more gracile frames. Our relatively lighter bones may make us more adept at long-distance running compared to sprinting, which could have allowed us to outlast faster-moving prey.

The other possibility involves our magnificent brains. GAGs do more than just help cushion our cartilage, they’re also expressed in neural tissue, stabilizing connections, locking circuits in place, and preventing any rewiring. A global reduction in GAGs throughout our bodies may have made our brains more neuroplastic and better able to learn and adapt.

In other words, an investment in our information sponges may have come at the cost of our joint sponges. It’s something to think about while you’re icing your knees.

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NAUTILUS

Ice Age Spear Points Tell a Fascinating Story of Human Progress

JAKE CURRIE · NAUTILUS · SEP 29, 2026

Archaeology essay on what tool-making choices reveal about early humans.

I imagine you’re suddenly transported to the wilderness, hundreds of miles from civilization, with nothing other than what you’re carrying with you right now. The way you view your personal effects would likely change drastically. With no cell towers or electricity, your cherished phone is useless, the money in your wallet is only good for fire tinder, but the drawstring on your sweatpants could be used as a fishing line, or maybe a snare to catch dinner.

The value we place on things is contingent on our circumstances and subject to change as they do. Anthropologists from the University of Mississippi and the University of Alabama recently demonstrated this in a study of stone spear tips belonging to two groups of hunter-gatherers who settled in the Tennessee River Valley during different time periods.

Read more: “The Emergence of a Deadly Hunting Technology in Prehistoric North America”

The Clovis people were the first humans to settle in the region around 13,200 years ago. As strangers in a strange land, it’s likely they were

more uncertain where to source the sedimentary rock they needed to fashion weapons. That’s probably why their stone spear tips showed signs of resharpening, the researchers say.

“We hang on to stuff today, but when we’re talking about mobile hunter-gatherers, they had a different understanding and worldview when it came to personal possessions,” study co-author Jesse Tune of the University of Mississippi said in a statement. “Spear points were designed to be disposable objects, and that makes what the Clovis were doing interesting.”

Several hundred years later, the Cumberland people settled in the same area and their stone spear points told a different story. Instead of resharpened tips, the researchers found usable spear points that had been discarded.

So what made the Cumberland people relatively more wasteful? According to the team, they likely learned where the rock came from their Clovis forebears, which made carrying spear points around less valuable.

“When you think about a people who are moving around a lot, understand that they didn’t have any pack animals,” study co-author Shane Miller of the University of Alabama explained. “You really don’t want to be carrying around a bag of rocks, and so you’re going to be judicious about balancing what you need versus what you can carry.”

Fast forward several thousand years and as we’ve pushed the frontiers of exploration, we’ve left disposable tools our ancient ancestors couldn’t dream of in places they never imagined visiting. In

fact, there are six high-tech landers gathering moon dust on the lunar surface right now.

It’s something that’s not lost on the researchers. “There really is no difference between Ice Age hunter-gatherers and the people sending missions and rovers to Mars,” Tune said. “We’re moving into an area, deciding what to take with us, learning the resources and slowly expanding. It’s the exact same process.”

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Lead Image: Kevin Bain/Ole Miss Digital Imaging Services

NAUTILUS

Do Transplanted Hearts Assume the Age of Their New Host?

KRISTEN FRENCH · NAUTILUS · SEP 29, 2026

Reporting on transplant research with implications for organ availability.

The miracles of modern transplant surgery can still sometimes boggle the mind. That the beating heart of one person can be sewn into the living body of another to save their life can seem like a Frankensteinian feat. But today, this kind of surgery is a standard and established medical procedure rather than an experimental one: In 2025, in the United States alone, more than 4,500 hearts were transplanted, according to the American Heart Association.

The tragedy is that it’s not available to everyone. Many thousands of people snake through long organ donation lists waiting their turn, and many of them die before their number comes up. That’s partly because everyone wants to receive a young, healthy heart rather than an old and weary one—the younger the better. This could soon change, if recent findings about how organs age hold up.

Read more: “If Only Technology Had a Heart”

The desire for young organs has been partly driven by a speculative theory of rejuvenation: Research in mice known as heterochronic parabiosis, where an older and younger mouse’s circulatory systems are sewn together, suggested that the blood of a younger mouse could reverse age in an old one. Some hypothesized that something similar might happen with organ transplants.

But a team of scientists and surgeons from Harvard Medical School, together with a surgeon from Berlin, recently discovered that instead, an organ’s biological age can change to match that of its host after transplantation. When they sewed an old heart into a younger mouse or human, the organ seemed to turn back time, growing younger in its new dwelling, due to factors in the molecular environment.

The reverse was also true, however: When a younger heart was surgically transplanted into an older mouse or person, the young heart aged to match the body of its recipient. The organs “assimilate,” the scientists proposed, like a person who moves to a new place and learns the local language and customs so they can fit in.

In mid September, the team published their paper as a preprint, meaning it has not yet been peer reviewed, so the work still needs to be evaluated and validated—but the findings could have implications both for transplant surgery and the science of aging and rejuvenation. I spoke with coauthor Jesse Poganik, whose lab at Harvard Medical

School studies the biology of aging, about the future of transplant surgery, cell senescence, and epigenetic clocks.

Was this finding something you expected when you started out?

Yes and no. Prior to this project, we were working with a system called heterochronic parabiosis, where you surgically join the circulatory systems of a young animal and an old animal to see how this affects tissue regeneration. It has been particularly popular in the aging field. We’ve shown, using epigenetic clocks, that in that model, the old mouse becomes biologically younger. We also know that the old mouse lives longer compared to controls after you detach the mice. And the young mouse becomes reversibly biologically older. Because of these experiments, in a sense, it was not surprising that we found when an old heart was put into a young organism, it became biologically younger or vice versa.

What was a little bit surprising was that in parabiosis, we see a very strong reciprocal effect. So old becomes young, young becomes old. Pretty much in every organ we’ve profiled, that’s true. Here, we didn’t see that. We were suspecting that we might see some reciprocal effect, some feedback onto the recipient organism—that a young heart might have made the older animal somewhat younger. But when we looked at the other organs of the recipient animals, we didn’t see any evidence of that. Young mice that received old hearts, based on the epigenetic clocks, did not become biologically older. Older mice who received younger hearts did not become biologically younger. But the graft itself, the transplanted heart, that’s where really strong effects were observed.

Were you able to look at systemic aging in the human patients who received transplants, as well?

Yes and no, again. On the one hand, we did look at baseline health indicators, and then also cardiovascular-specific ones. The difference is that, in the human case, it’s not only hugely more complicated. We used a syngeneic mouse model, meaning the two mice are inbred so that they have similar immune systems. They don’t need immunosuppression. They’re also not, for a lack of a kinder way of saying this, on the verge of death before they get the transplant.

For humans, of course, a heart transplant is a last resort. Because of the realities of waiting lists and so forth, many transplant recipients, particularly heart transplant recipients, are extremely sick by the time

they get their new hearts. In other words, it's not exactly one-to-one. The mouse model is also a heterotopic model, meaning there is actually a second heart that is implanted in the mouse. The first heart also remains. And that turns out to be an established model of heart-transplanted mice. It doesn't recapitulate every single thing that goes on in a heart transplant in humans, but it turns out to be, from an experimental point of view, very helpful because the native heart that the mouse retains is a very useful control for us. When we looked at the feedback into the systemic environment, one of the things we



Continued online (full article about 2346 words). Scan, or visit <https://nautil.us/do-transplanted-hearts-assume-the-age-of-their-new-host-1285381/>

NAUTILUS

Aging Muscle Has a Hidden Molecular Switch

JAKE CURRIE · NAUTILUS · SEP 29, 2026

Research write-up on the biology behind muscle aging.

When it comes to the skeletal muscle we use to move around, there are two general types: fast twitch and slow twitch. We use the former, which rely more on simple carbohydrates for fuel, for quick motions—sprinting, jumping, snagging an errant toddler before they collide with the corner of a coffee table, etc. Slow-twitch fibers, on the other hand, are packed full of mitochondria and used for endurance activities like jogging, maintaining our posture, and hauling toddlers around when they run out of energy.

As you might imagine, our reliance on the two types changes over time. As we age, our bodies shift from fast-twitch muscle fibers to slow-twitch muscle fibers, which presents something of a paradox. Our mitochondria also lose a bit of their pep as we get older, so why do we increasingly rely on the muscle fibers that depend on them?

Read more: “Can a Drug Reverse Age-Related Muscle Weakness?”

To find out, a team of metabolic researchers looked at a molecule that's only present in the tiny cellular powerhouses: cardiolipin. Cardiolipin is found exclusively in the inner membrane of mitochondria and helps give it its signature folded structure. As we age (or if we develop a metabolic muscle wasting disease), cardiolipin levels decline, which is something that also happens in mice. So the researchers artificially reduced cardiolipin in young rodents and discovered the same age-related shift from fast- to slow-twitch fibers. In other words, the decline in cardiolipin seems to be causing the twitch switch.

looked at was the native heart, and we didn't see any effects on the aging of the native heart.

To answer your original question about the humans, it's a little hard to say because baseline health is generally improving anyway, simply because the people are so sick by the time they get the transplant. And the transplant usually turns out to be transformational in most cases for most recipients. The expectation is that the health improves in general.

Is there any sort of limit in terms of the viable difference in age between the donors and recipients of the transplants?

According to the team's new paper published in *Nature Aging*, it's a defense mechanism. Cardiolipin-depleted mitochondria produce more reactive oxygen species (oxygen molecules with unpaired electrons that can cause cell damage). When the researchers used antioxidants to mitigate these reactive oxygen molecules in muscle cells, they also slowed the shift toward slow-twitch fibers.

“The fiber switch is not the muscle failing, but the muscle trading power for protection,” study author Fabian Finger of the University of Copenhagen explained in a statement. “That's also why interfering with it can backfire: when we gave the mice antioxidants to mop up the ROS, their muscles fared worse, not better.”

They also found that the protein ERRγ (estrogen-related receptor gamma) acted as a key mediator of the process. Once this nuclear receptor receives a signal from the reactive oxygen molecules, it triggers the muscle cells to switch to slow twitch. In cultured muscle cells with no ERRγ, no switch occurred.

The team believes these findings in mouse models could point the way toward therapies in humans. “What encourages me most is that even a partial recovery of cardiolipin was enough to bring the muscle back,” study author Zachary Gerhart-Hines of the University of Copenhagen said. “The question now is whether we can increase cardiolipin in aging muscle or target ERRγ to promote healthy adaptations. This is where the therapeutic potential lies.”

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PSYCHE

The full-service grandad

LIAM HENEGHAN · PSYCHE.CO · SEP 30, 2026

Personal essay on fatherhood and grandfatherhood worth the full read.

The grandmother hypothesis is well known. Grandmothers are said to improve evolutionary fitness, helping children survive and families endure across generations. I know a grandma who carries a spare diaper in her handbag. Then there's the grandfather effect. According to some anthropologists, grandads reduce the survivorship of offspring. When a grandad mooches around, his sons have fewer kids. We snooze, we're odiferous, we eat too much – right out of the mouths of babes! But if science has been slow to celebrate us, life has recently handed me a counterargument. Three years ago, when my granddaughter came home from the hospital, I committed myself to becoming what I now call a 'full-service grandad'.

I too had grandfathers. One, called Danda, daily consumed fried bread and loved his rashers of bacon. He would run with me up and down the length of Xavier Avenue in Dublin's North Strand, and when the trains pulled out of the station, we paused to wave, up, up, up so high on the bridge at the end of that short street. The other grandfather, Grandy, smoked his pipe, or rather he prepared his pipe, shaving the plug of tobacco, stuffing the bowl, striking the match, and raising it aloft as he talked till it burned down to his fingers. His background was revolutionary, storyful, pious; the memories linger a half-century after he is gone.

Even before I became a grandfather, I looked the part. My hair greyed in my 30s and then whitened; my beard is billowy and snowy. My complexion, though not quite rosy, is a little pinkish. December is a cruel month for men embodied in this way. This year, a young man crossed over from his line to mine in Trader Joe's to announce that I look like Santa. It was the 7th of January. 'Christmas is over, friend,' I said. 'I've reverted to being Karl Marx.' He shrugged, unrepentant, as they always are. But in the main, I look like a generic grandad. Even before my granddaughter was born, children glanced up at me with timid recognition. My granddaughter is now three, and I have grown into the role. Despite ongoing requests from her grandmother – Yiayia – I won't cut my hair or shave: 'Why upset the child?' By genetics, by age and by a natural joviality, I am a fully gestated grandfather.

When my granddaughter was bundled up and readied for her homecoming from the hospital to her first apartment in Lincoln Square in Chicago, I travelled behind the main party by train and walked the chilly street from Ravenswood to join the family. It is a route I've walked hundreds of times since. I had committed to being a full-service grandad. The offer had been made and accepted: I was to be a caretaker for two full days a week for as long as her parents needed me. I had, I think, been a very present dad to our two boys. And yet, years before my granddaughter was born, I expressed to the family – everyone was home for the holidays – 'Wasn't it great that, because I'm a professor, I could be home so much when you were small?' How everyone laughed that day. When the chortling subsided, a son had mopped his eyes, he said: 'You were never around.' Had I missed out, I wondered? Maybe. One thing I do know about family life now: despite how arduous and all-consuming it is, no matter how tired you feel, that phase is over too quickly. I am a full-service grandad, and yet I realise there is a clock ticking on that service.

Does a full-service grandad change diapers? Yes, he does. This is perhaps the first question a prospective full-service grandad might ask. I recall that this delicate question was on my mind when, just a

few weeks after she was born and my 'training day' was complete, it was just her, me, and a really quite unpleasant deposit. Was I really up for this, I wondered. But yes, it's part of the deal. I have since changed – by my back-of-the-envelope calculation – 686 diapers and then pull-ups over the past three years. The child is now potty trained, but I sometimes glance at the last of the pull-ups on her bedside table and wonder when a diaper might have my name on it – a geriatric question for another essay. So much for diapers. But just as importantly, I have overseen feedings, nap times, tummy time, library visits, stroller jaunts and, above all, park visits, playground romps, river walks, bird-watching, squirrel-noticing, beach days, and a full immersion in nature. This is what I am here to say: the full-service grandad does all the routine labours of care, but must also ask what distinctive contributions he brings.

There is something incomparable about offering what you have while they are still so young

I had promised myself that I would not turn my granddaughter into a project. It is tempting, as an ecologist, to observe her too closely. But what interests me most is not what she reveals as a subject, but what she helps me see again. Yes, yes – that is a sparrow, a doggie, a worm, a hawk in a tree, a puddle worth splashing in. The success of my method lies in the fact that young children and elderly gentlemen alike take everything slowly. I will not make a study of her, and yet by catering to her attention I've learned how to see the world with fresh wonder.

Each full-service grandad must discover his own talent. Mine is to immerse my granddaughter in the green and leafy world – an intense exposure she might not otherwise have had, at least not so early. Who knows what the future holds for this tiny child who has braced herself against Chicago's bitter winter days, leapt wellington-booted into spring-swollen puddles, and ambled along sunbaked sidewalks with a floppy hat pulled low? She knows the moored boats along her stretch of river and pauses to remark on the geese drifting across North Pond. She knows her goldenrods, her butterflies, her asters. She never fails to notice the sky. The urban outdoors will have left its mark. And each grandad brings his own gifts: familiarity with the beasts of the countryside, an eye for the stars, a mechanical mind, a musician's ear, or a facility with art. Imparting these gifts is not the work of an idle afternoon; they require time, patience, and presence. Grandads in the traditional mode may well shine later in a child's life, but there is something incomparable about offering what you have while they are still so young. Committing to being a full-service grandad begins with a simple act of self-interrogation: what part of the world can I open for this child?

I am a full-service grandad, but it's a lonely task. I am often the solo grandad in the library, the park, and at the beach. There is, of course, a well-documented crisis in childcare: it is expensive for a parent to stay at home, expensive to send children to daycare, and expensive to hire a carer. And though grandparents often step up – my granddaughter is in the exceptional care of my daughter-in-law's mother two days a week, and Yiayia, more often than not, shares the responsibility with me now that she is retired – even those contributions come at a cost. We are all finite beings, of course, but finitude feels more urgent in one's 60s than when one is 20: time engaged in one pursuit means less time for others. I still work full-time, so time not spent with her is fully consumed by paid work. I

have written less, applied for fewer grants, and postponed conference-going. My years with my granddaughter have cost the world at least one book and two or three essays. That's at least a few hundred people deprived of desultorily glancing over my prose. It is both a cost and a very great privilege to do the work of a full-service grandad but, gendered norms being what they are, very few of us are availing ourselves of this opportunity. I want to see more grandads in the fray. Let us nod to one another as the kids throw a hundred pebbles into Lake Michigan – a slight salute in the Nature Museum's butterfly haven. We have been an evolutionary mixed bag for far too long; it's time to redress the historical imbalance.

On creaky legs, hoist yourself from your easy chair, Grandad. Apply a final touch of pomade to thinning hair and place that hat firmly upon your head. We shall skip stones on the lakefront; place block upon block in the children's corner of the library; push strollers through the streets and encourage toddling steps upon the grassy hills of our parks. We shall never surrender that little hand when crossing a busy street. There is no crisis of loneliness in the Brotherhood of Full-Service Grandads, only a surging confidence in our mission. Grandads of the world, we have done well – but now we must do better. We have nothing but our naps to lose, and a world to set right: a little child to care for.

NAUTILUS

The Sleep Cycle of Ideas, Seeded Dream Poems, and the Queen of Dinosaurs: 8 Books We're Excited to Read This October

BOB GRANT · NAUTILUS · SEP 29, 2026

Nautilus's October reading list, lighter but still substantive.

As summer wanes, we're entering curl-up-with-a-good-book-in-a-cozy-nook season. Bring it on, I say. All the better if it's a nice, meaty title that gives your grey matter something to chew over. These books, hitting shelves soon, promise to provide just such stimulation, comfort, and challenge.

Tyrant Lizard Queen: The Love, Life, and Terror of Earth's Greatest Carnivore by Riley Black

Recently, researchers have produced a plethora of insight into *T. rex*—the dinosaur species that stands alone as the spokesreptile of that once mighty branch of Earth's tree of life. This year alone, we've learned why *T. rex*es had tiny arms, that they had body temperatures similar to humans (97 Fahrenheit), the walking speed of at least one adult *T. rex* (about 3.5 to 4.5 miles-per-hour), and that the massive beasts took from 35-40 years to grow to full, adult size. What a perfect time for award-winning science writer Riley Black to lead an evidence-based hike through the life of fictitious *T. rex*, Snaggletooth. In *Tyrant Lizard Queen*, Black follows Snaggletooth as she hatches, survives, plays, sleeps, mates, and nests in Cretaceous North America. Sign me up for this walk in the woods.

Why We Ask Why: The Science of Explanation and the Human Drive to Understand by Tania Lombrozo

Anyone who's spent time with a toddler or a scientist knows that "whys" come fast and furious. One might argue that "Why?" is the singular question that separates humans from our fellow Earthlings. Princeton University psychologist Tania Lombrozo has tackled big questions on *Nautilus* before—how to attain existential comfort without religion and the diceyness of "playing God." In her latest book, *Why We Ask Why*, she takes on the mother of all questions, asking why humans so often ask "Why?" With scientific and philosophical insight aplenty, this one seems like quite the read to wrap our questioning minds around.

The Taste of Rain in Buenos Aires by William Dowd

We at *Nautilus* have been waiting for this one for a few years. Our very own Kristen French talked with MIT-trained poet and science writer William Dowd in 2023 about his audacious project to

recapture the wonder of reading—stolen from him through a condition called binocular vision disorder—by trying to seed his own dreams. With the help of an experimental dream inception device called *Dormio*, Dowd used unfamiliar poetry to try and influence the content of his nightly vision quests. *The Taste of Rain in Buenos Aires* details the results of this yearlong experiment. Each of Dowd's 20 guided dreams metamorphoses into a prose poem on the page.

42 Tales from Life: A Modern Bestiary by Richard Dawkins

Fifty years after the publication of his most famous book, *The Selfish Gene*, famed evolutionary biologist Richard Dawkins adds to his impressive bibliography with what seems to be a less-loaded topic. In *42 Tales from Life*, Dawkins presents a celebration of the evolution of animals where the protagonists are life's extraordinary adaptations—the most intriguing physical traits and behaviors in Earth's menagerie, as shaped by the creative process. The 42 vignettes are accompanied by the breathtaking art of Ernst Haeckel, the 19th-century polymath whose botanical and zoological artwork has delighted viewers (including Dawkins) for decades.

On the Equality of All Things: Lessons on Physics and Philosophy by Carlo Rovelli

Unprecedented leaps in humanity's understanding of the universe demand clear-eyed explanation. As science peels back more and more layers of our reality, theoretical physicist Carlo Rovelli has long been happy to oblige—to describe what we're left with. His latest book, *On the Equality of All Things*, helps readers navigate our current era of perspective-shifting discovery by weaving together modern scientific insight and ancient philosophy. The nature of time, the illusion of cause and effect, the fragility of reality. Nothing is off limits.

Sleeping Beauties: Why Good Ideas Go Dormant and How They Wake Up by David Byrne

Same as it ever was. From iconic lyricist and Talking Heads frontman David Byrne, comes a new meditation on the life cycle of ideas. *Sleeping Beauties* charts the stuttering course that groundbreaking innovations took from ideation to actually breaking ground.

Throughout history, artists, thinkers, scientists, and inventors have been stymied by circumstance, their transformative ideas forced to hibernate. But these same ideas often persevere and percolate, later reawaking to change our technologies or our views on the world. Byrne illustrates several such paths and offers a lens through which we might avoid sleeping on the next big thing.

Wireheads: An Unnatural History of Technology in the Brain by Mike Jay

Science has so much yet to learn about the brain. But as neuroscientific insights continue to pour in, we've not been shy about using technology to interact with the complex organ. Science historian Mike Jay, author of 2023's *Psychonauts*, probes the intersection of brains and technology in his latest work, *Wireheads*. From mind-controlled raging bulls and pleasure-overloaded rats to

CIA brainwashing and Elon Musk's Neuralink, Jay explores the promise and peril of marrying brains with machines.

Worlds Unfolding: Science on the Page by Jennifer A. Watts, Daniel Lewis, Joel A. Klein, and Natalie Lawler

This five-volume set is a distillation of the Huntington Library's collections on the history of science, medicine, and technology. It looks to be a fascinating compendium of human knowledge-seeking, cleverly segmented into neat categories: Above (astronomy, aviation, aerospace); Below (geology, mining, oceanography); Within (anatomy, medicine, electricity); Beyond (exploration, science fiction); and Alive (natural history, ecosystems). Add to these a booklet that includes an introduction by none other than Bill Nye, science guy, and these volumes could provide hours and hours of cozy reading.

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